

Mobile Devices and Applications

UNIT 2 – ANDROID BASICS

Agenda

Objectives

Content

Evaluation

Resources

Objectives

Efficient usage of mobile technologies in the information society context.

Understanding the principles of mobile devices.

Learning and achieving Android development skills.

Evaluation

Course activity: 50%

- Final quiz exam – 50%

Laboratory activity: 50%

- 2 Partial Evaluations – 30%
- Ongoing evaluation – 20%

Final evaluation: January 21st, 2026

70% seminar attendance is mandatory

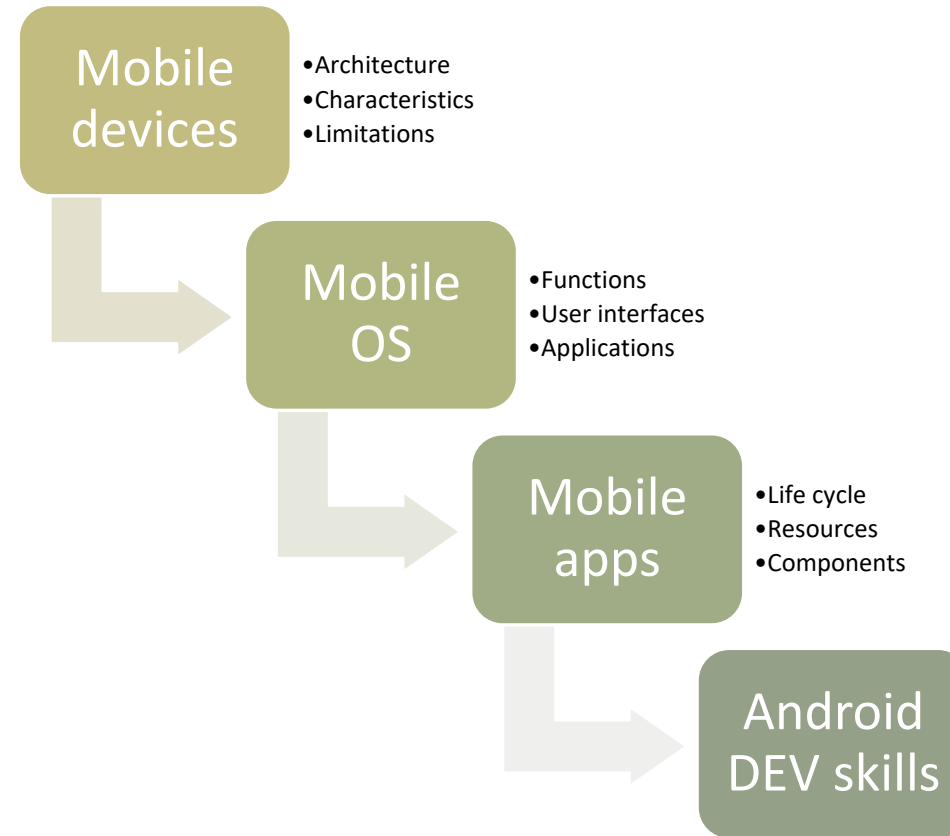
All evaluation activities must be passed with a minimum of 50% from the total number of points

<http://pdm.ase.ro>

Agenda

1. Mobile devices
2. Android framework
3. Development instruments
4. Android application structure

Learning path



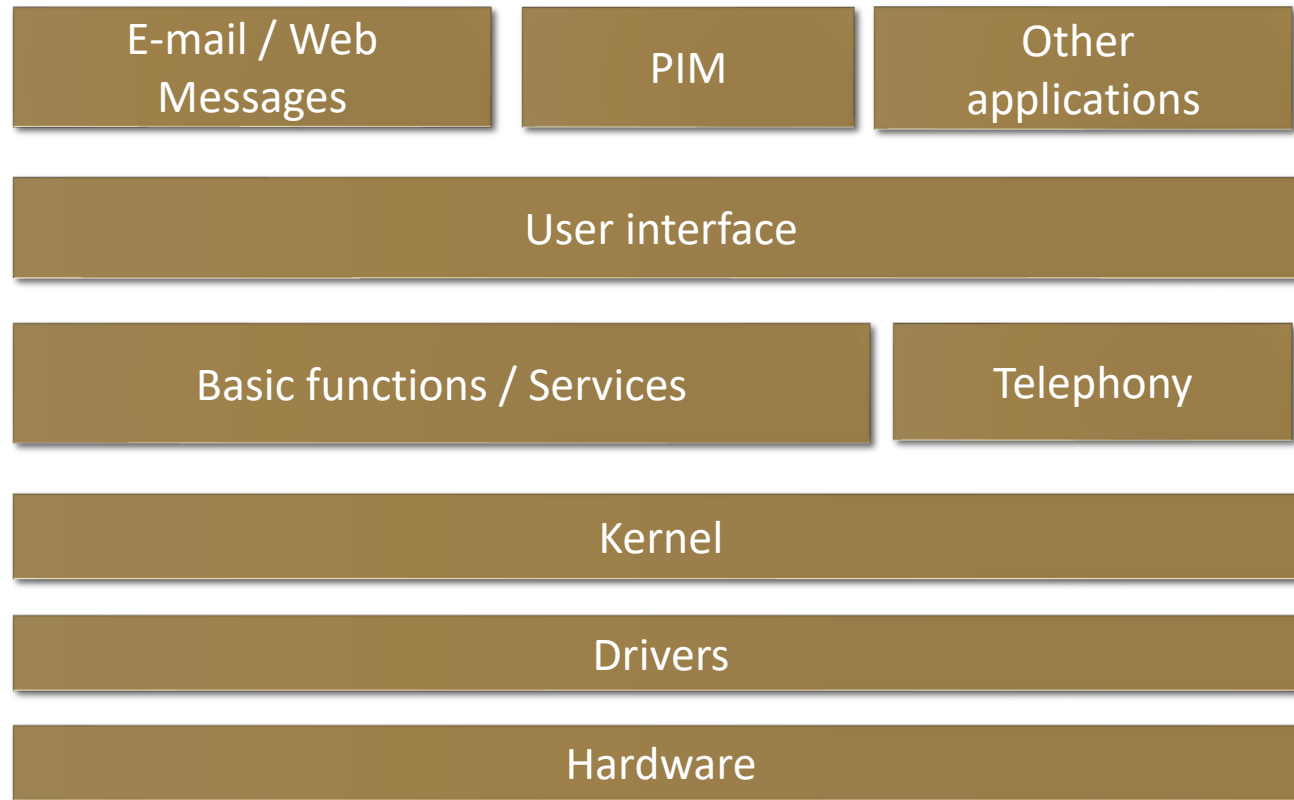
Mobile operating systems

ARCHITECTURE AND MAIN FUNCTIONS

Mobile Operating Systems

- Manage the hardware and software resources
- Manage the memory
- Control the processes
- Check the I/O gates
- Run applications
- File management
- Check the user interface

Mobile Operating Systems



Mobile operating systems

Android (Google)

Bada (Samsung)

Tizen (Tizen Association)

BlackBerry OS (BlackBerry/RIM)

BREW (Qualcomm)

Firefox OS (Mozilla)

iOS (Apple)

Linux Mobile

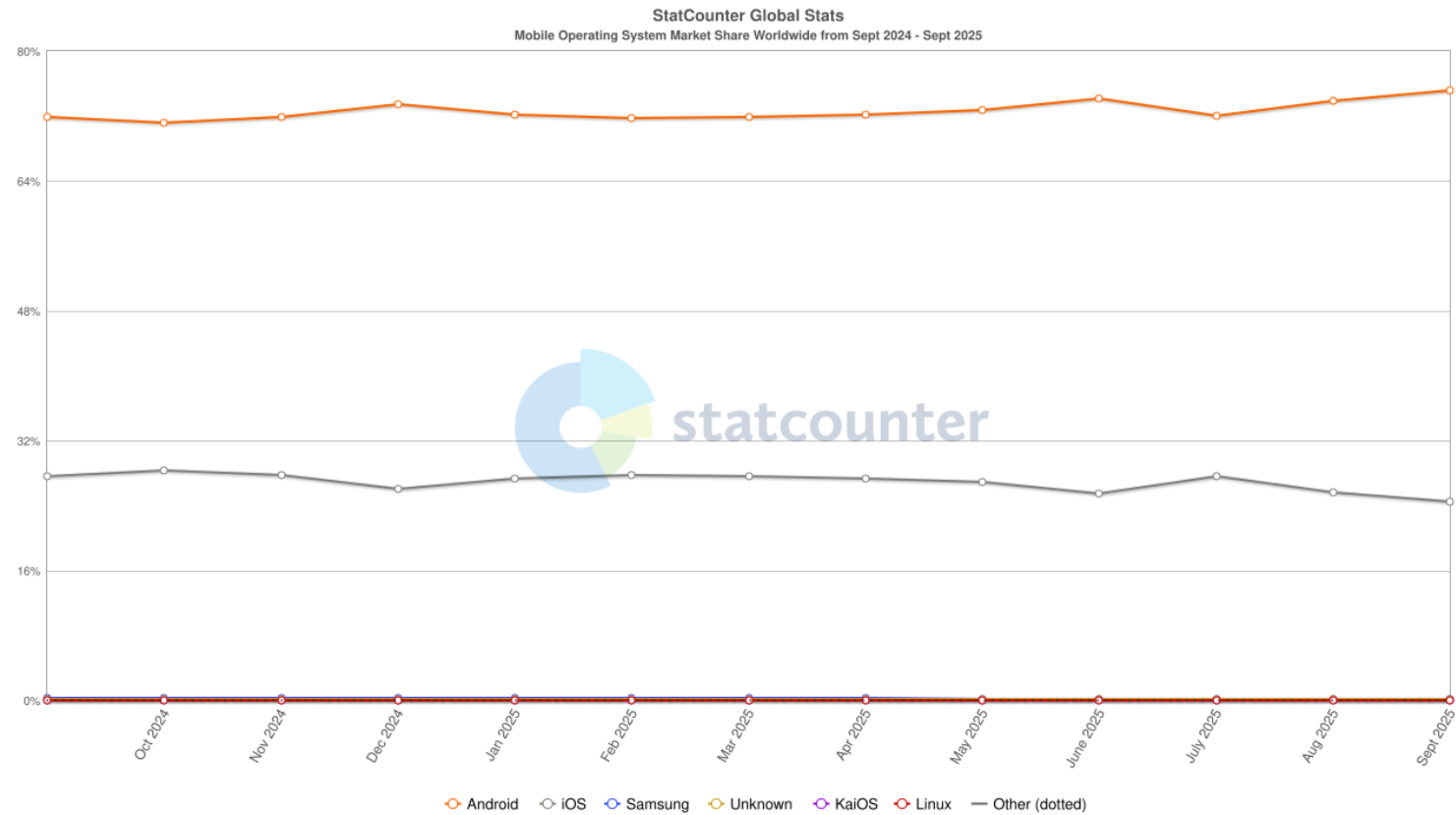
Palm OS/Garnet OS (Palm)

Symbian (Nokia) – obsolete

webOS (HP)

Windows Phone/Windows CE/Windows Mobile (Microsoft) – obsolete

Smartphone 2020



Android

Bought by Google in 2005

Open Handset Alliance in 2007

Powered by Linux 2.6.x/3.x kernel

Java/Kotlin based applications

Phones

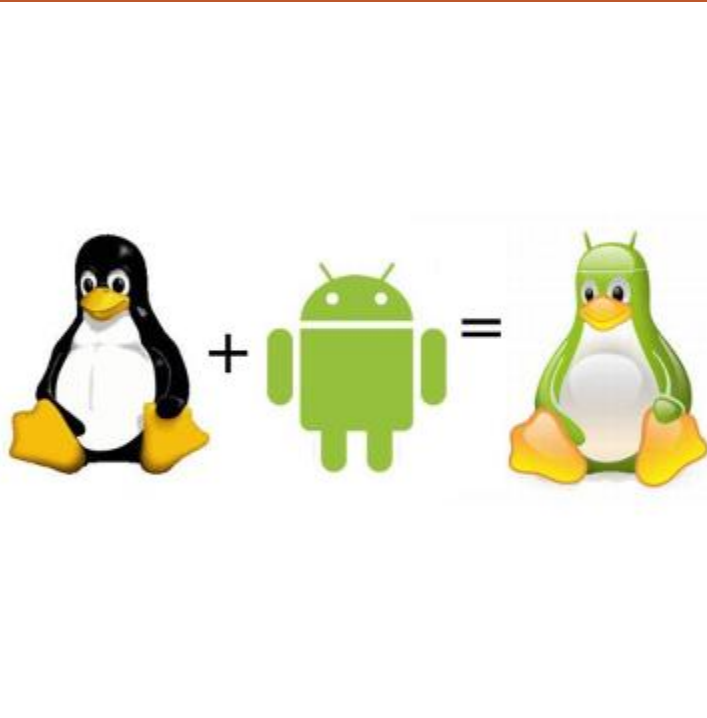
- HTC One M8, Samsung Galaxy S5, Google Nexus 5

Tablets

- Samsung Galaxy Tab, Google Nexus 7

Android

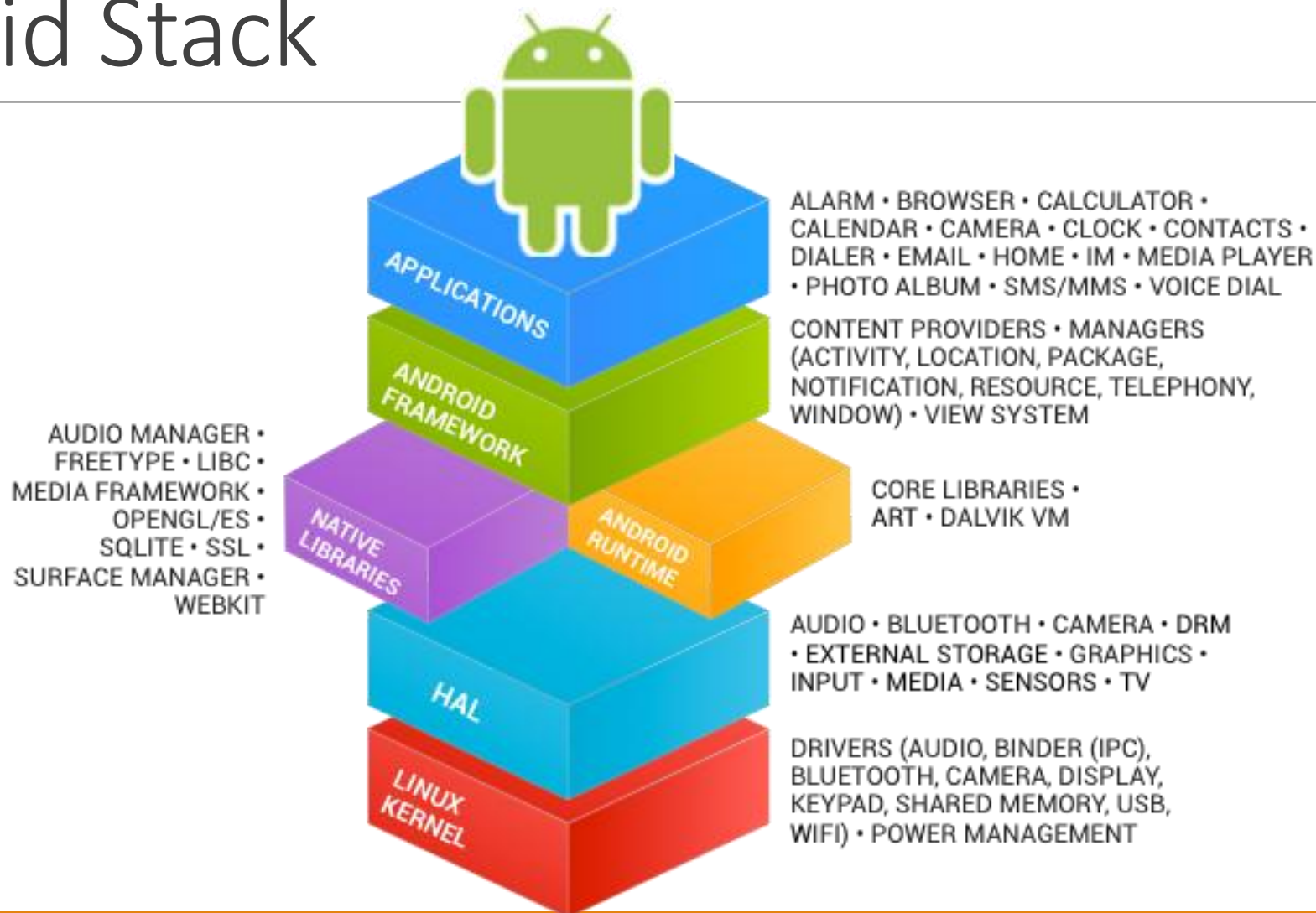




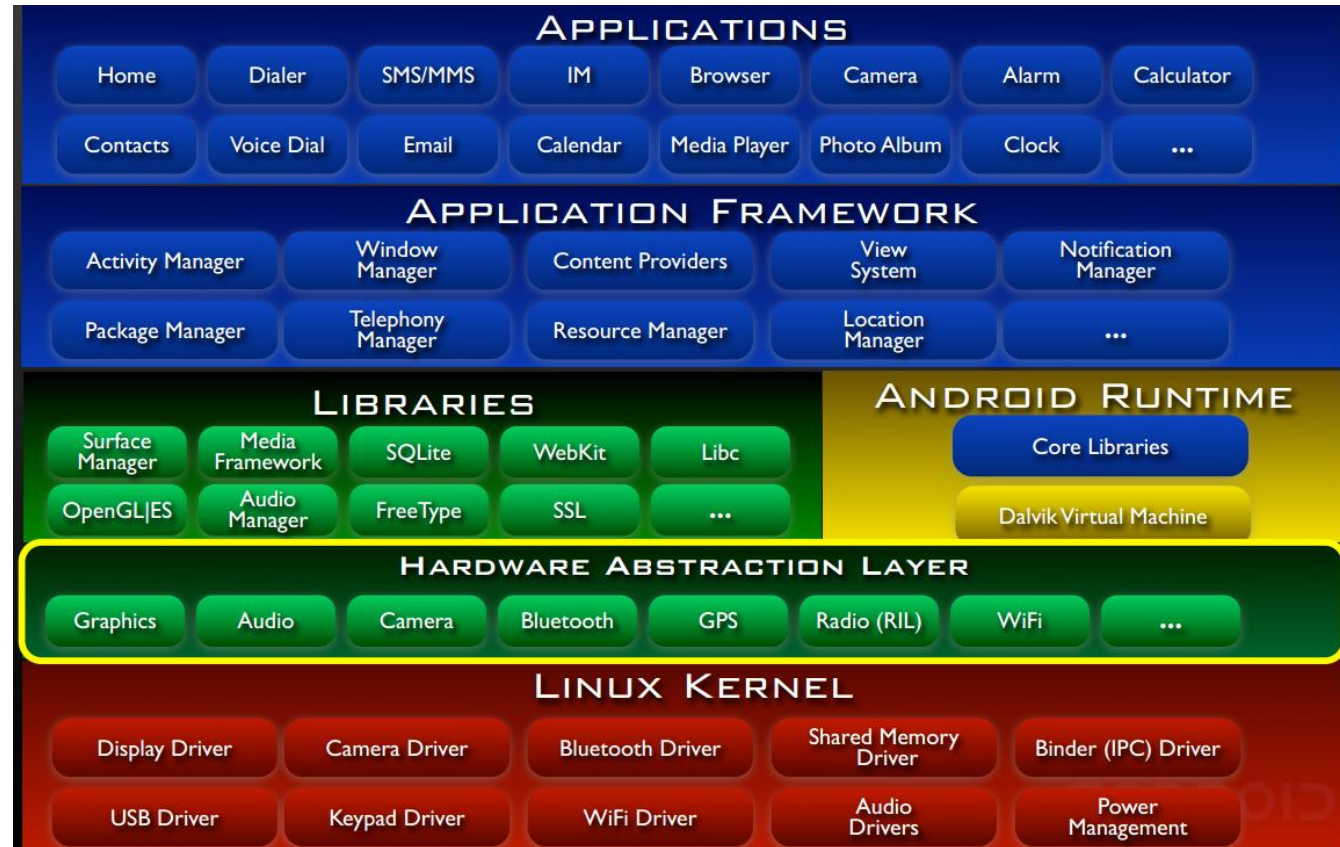
Free and open source software

Sailfish OS, Tizen, Ubuntu Touch OS, OpenMoko Linux, Fedora,
CentOS, Debian, SUSE, RedHat

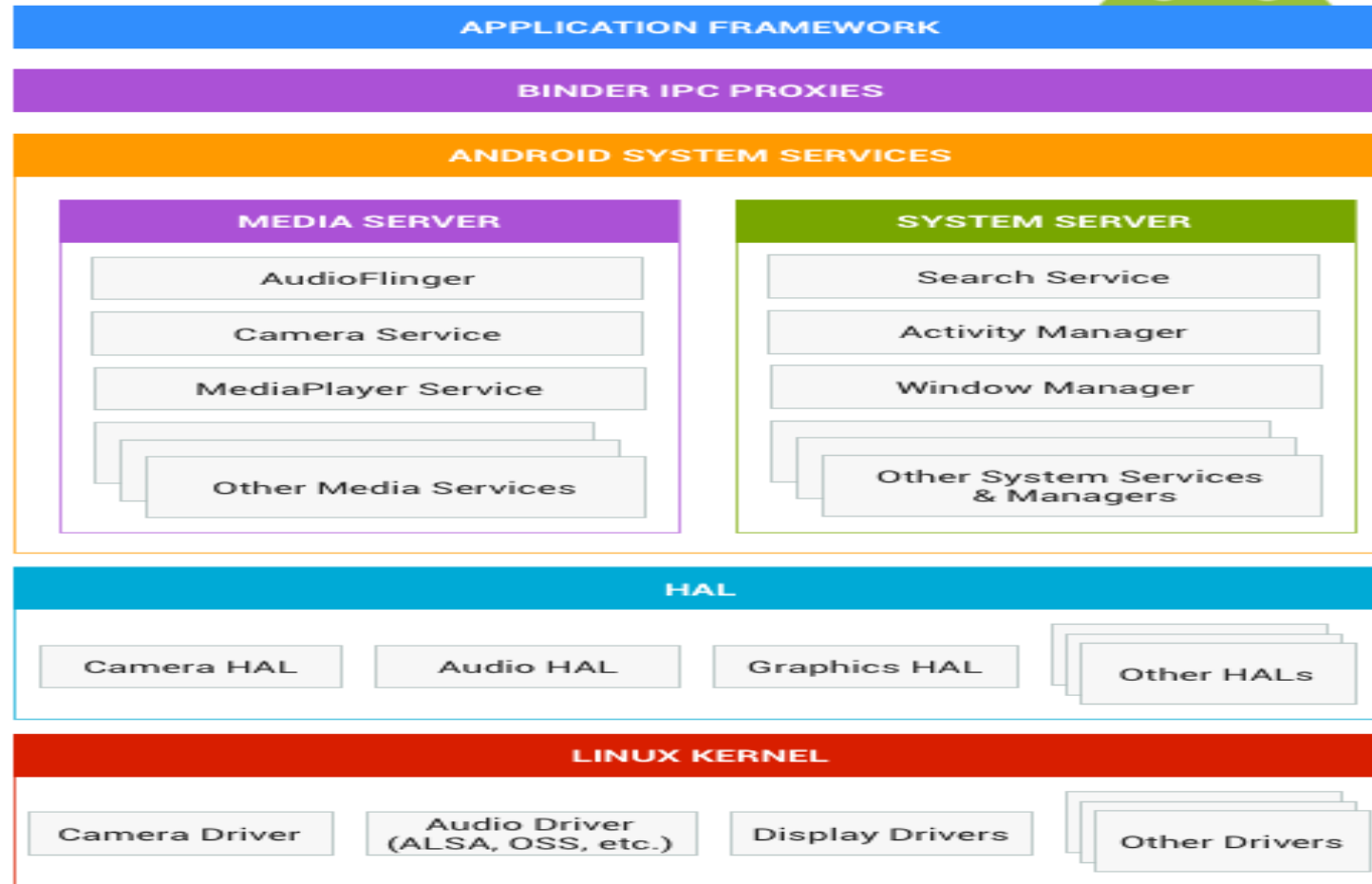
Android Stack



Android Architecture



Android Architecture



Dalvik Virtual Machine

Android's custom clean-room implementation virtual machine:

Provides application portability and runtime consistency

Runs optimized file format (.dex) and Dalvik bytecode

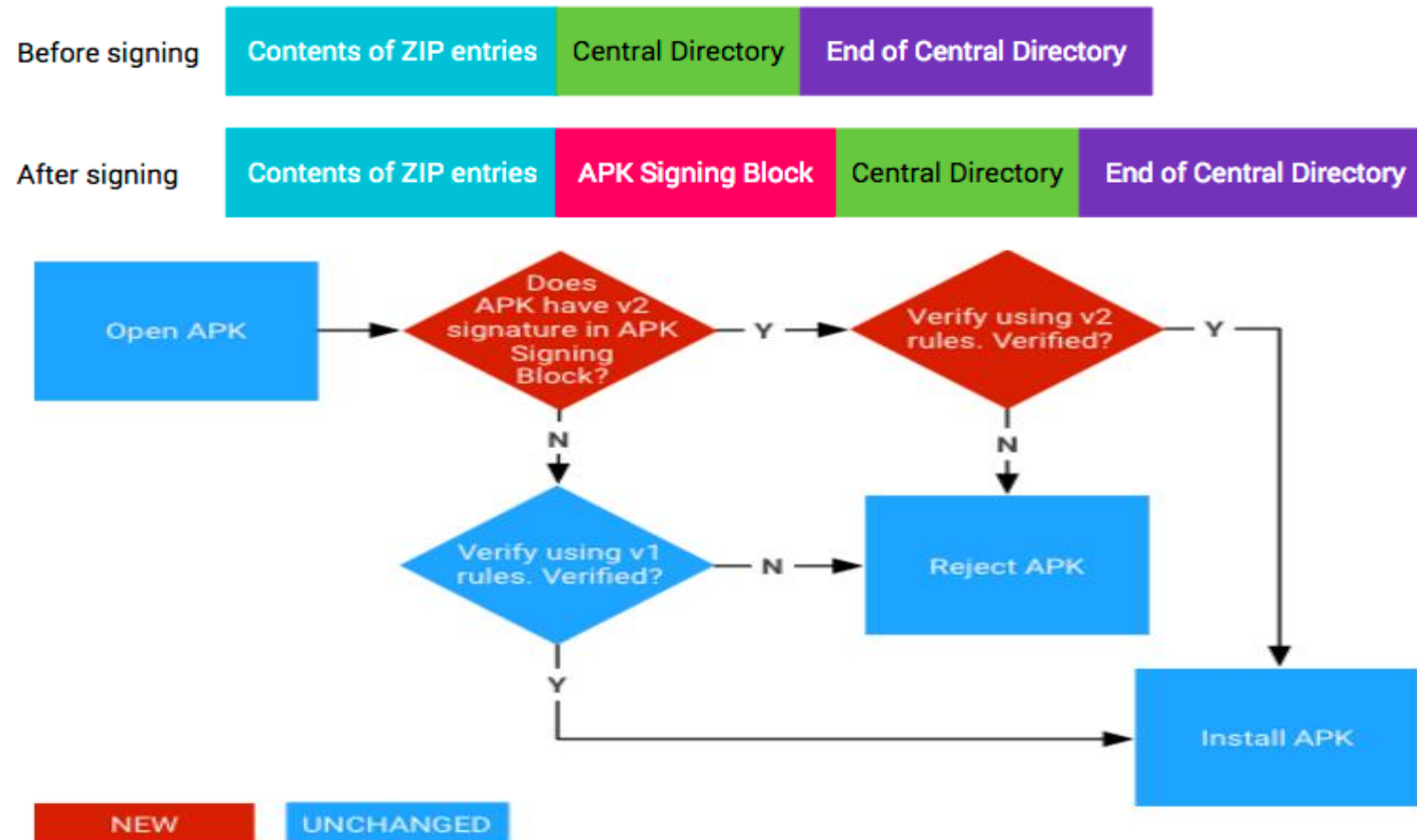
Java .class / .jar files converted to .dex at build time



Mobile Applications

ANDROID APP STRUCTURE

APK Signature Scheme



Android Binary Files

APK extension (Android Package)

An archive with the optimized Dalvik code and resources

Resources can be either compiled or not

Besides resources any other file can be added

Components

Sources (**src**)

- Java source files(**java**)
- Resources(**res**)
 - res/drawable
 - res/layout
 - res/values
 - res/menu
 - res/xml
 - res/raw

Resources obtained from files (**assets**)

Configuration file (**AndroidManifest.xml**)

Compiling properties

- **build.gradle**

Generate files (**gen**)

- **R.java**

AndroidManifest.xml

Package information (name, version)

Application attributes (name, icon, theme, memory options, restrictions, permissions, etc.)

Message filters used for app components

Uses permissions

- **<uses-permission** android:name= "name"/>

Hardware and software requirements

- **<uses-feature** android:name= "name" android:required="true/false"/>

App's components

- Activity declaration, services, content providers, message receivers
- The class names associated with app components
- Properties

SDK versions (minimum, maximum, desired)

- Exposed in *build.gradle*

Permission examples

Scope	Android permission
Internet/Network access	INTERNET
Contact READ/WRITE	READ_CONTACTS, WRITE_CONTACTS
Calendar READ/WRITE	READ_CALENDAR, WRITE_CALENDAR
SEND/READ/WRITE SMSs	SEND_SMS, READ_SMS, WRITE_SMS
TELEPHONY ACCESS	CALL_PHONE
READ/WRITE on external storage devices	READ_EXTERNAL_STORAGE, WRITE_EXTERNAL_STORAGE
Determining the GPS coordinates	ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION

Hardware and software requirements

android.hardware.camera

android.hardware.camera.autofocus

android.hardware.camera.flash

android.hardware.nfc

android.hardware.sensor.gyroscope

android.hardware.Bluetooth

android.software.live_wallpaper

android.software.home_screen

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/android"

    package="ase.pdm.sem1"    android:versionCode="1"    android:versionName="1.0" >

    <uses-permission android:name="android.permission.INTERNET"/>

    <application android:allowBackup="true"

        android:icon="@drawable/ic_launcher"

        android:label="@string/app_name"

        android:theme="@style/AppTheme">

        <activity> ... </activity>

        <service>...</service>

        <provider>...</provider>

        <receiver>...</receiver>

    </application>

</manifest>
```

build.gradle

```
android {
    compileSdkVersion 23
    buildToolsVersion "22.0.1"

    defaultConfig {
        applicationId "ro.ase.pdm.myapplication"
        minSdkVersion 16
        targetSdkVersion 23
        versionCode 1
        versionName "1.0"
    }
    buildTypes {
        release {
            minifyEnabled false
            proguardFiles getDefaultProguardFile('proguard-android.txt'), 'proguard-rules.pro'
        }
    }
}

dependencies {
    compile fileTree(dir: 'libs', include: ['*.jar'])
    testCompile 'junit:junit:4.12'
    compile 'com.android.support:appcompat-v7:23.0.1'
    compile 'com.android.support:design:23.0.1'
}
```

Programming model

Linux based operating system

- C native libraries

Based on Java

Native programming interface

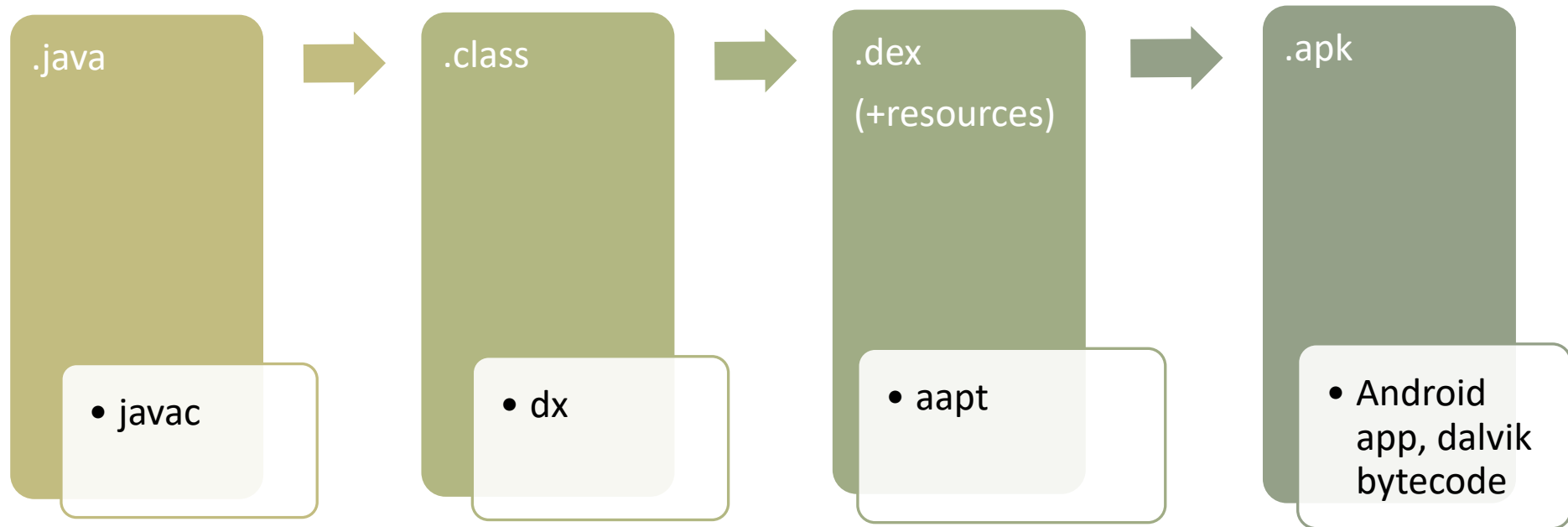
Personal virtual machine (**Dalvik VM**)

- Binary code not compatible with Java SE
 - Based on dex files
- Each application runs in a different process
- Using JIT compiler

ART – Android Runtime

- The current execution Android environment
 - Starting with Android 4.4
- Compile before execution

Android binary files



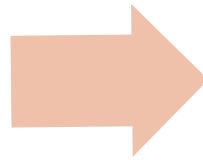
Android binary files

apk

- /META-INF
- /lib
- /res
- /assets
- AndroidManifest.xml
- classes.dex
- resources.arsc

Android binary files

dexopt – Dalvik
JIT – just in time
compilation



dex2oat – ART –
native executed
AOT –ahead of
time compiler

- .odex – optimized dex file

- elf – a compiled
dex file

Development instruments

Programming functionalities

User interface

Database

- SQLite

Media API

- Audio
- Video

Camera

Communication

- Socket
- HTTP
- Bluetooth
- NFC

2D Graphics, Animation

3D Graphics, OpenGL

Personal information

Telephony

Sensors

- Motion
 - Accelerometer
 - Gravity sensor
 - Gyroscope
 - Rotational vector sensors
- Environmental
 - Barometers – pressure
 - Photometers – illumination
 - Thermometers - temperature
- Position
 - Orientation sensors
 - Magnetometers - compass
 - Location

Development instruments

Software

- Java SE Development Kit (JDK)
- Android SDK
- Android Studio
- Android Tools

Integrated development environment

- Android Studio – Google official release
- Eclipse
 - plugin: Android Development Toolkit (ADT)
- IntelliJ IDEA

Android SDK



A set of specific internal libraries for Android



Resources and image emulators



Platform specific resources



Tools for compiling and binary code generation

Android SDK

Android SDK Tools

- Platform independent
- Subdirector: sdk/tools
- Instruments:
 - Ant scripts for generating the binary package
 - Debug monitor (ddms)
 - emulator-arm, emulator-x86

Android SDK Platform-tools

- Platform dependent
- Director: sdk/platform-tools
- Instruments:
 - adb – android devices communication
 - sqlite3

Android SDK Build-tools

- Director: sdk/build-tools/version/
- Instruments
 - aapt – resources compilation, R class generation, APK creation
 - dx – converting Java binary code to Dalvik binary code

Android SDK Manager

Instruments and platform management

Direct access or from the IDE interface

Content

- Libraries
- Source code
- Documentation
- Emulators

Android SDK Manager

Settings for New Projects

Appearance & Behavior

Appearance

Menus and Toolbars

System Settings

Passwords

HTTP Proxy

Data Sharing

Date Formats

Updates

Android SDK

Memory Settings

Notifications

Quick Lists

Path Variables

Keymap

Editor

Plugins

Build, Execution, Deployment

Kotlin

Tools

Appearance & Behavior > System Settings > Android SDK

Manager for the Android SDK and Tools used by Android Studio

Android SDK Location: C:\Users\mdoinea\AppData\Local\Android\Sdk [Edit](#) [Optimize disk space](#)

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, Android Studio will automatically check for updates. Check "show package details" to display individual SDK components.

	Name	API Level	Revision	Status
☐	Android 10.0+ (R)	30	3	Partially installed
☐	Android 10.0 (Q)	29	5	Not installed
☐	Android 9.0 (Pie)	28	6	Not installed
☐	Android 8.1 (Oreo)	27	3	Not installed
☐	Android 8.0 (Oreo)	26	2	Not installed
☐	Android 7.1.1 (Nougat)	25	3	Not installed
☐	Android 7.0 (Nougat)	24	2	Not installed
☐	Android 6.0 (Marshmallow)	23	3	Not installed
☐	Android 5.1 (Lollipop)	22	2	Not installed
☐	Android 5.0 (Lollipop)	21	2	Not installed
☐	Android 4.4W (KitKat Wear)	20	2	Not installed
☐	Android 4.4 (KitKat)	19	4	Not installed
☐	Android 4.3 (Jelly Bean)	18	3	Not installed
☐	Android 4.2 (Jelly Bean)	17	3	Not installed
☐	Android 4.1 (Jelly Bean)	16	5	Not installed
☐	Android 4.0.3 (IceCreamSandwich)	15	5	Not installed
☐	Android 4.0 (IceCreamSandwich)	14	4	Not installed
☐	Android 3.2 (Honeycomb)	13	1	Not installed
☐	Android 3.1 (Honeycomb)	12	3	Not installed

☒ Hide Obsolete Packages ☐ Show Package Details

OK

Cancel

Apply

Help

API Versions

Android OS	API Level
Android 1.5 (Cupcake)	3
Android 1.6 (Donut)	4
Android 2.0 – 2.1 (Éclair)	5 - 7
Android 2.2 (Froyo)	8
Android 2.3.3 – 2.3.7 (Gingerbread)	9 – 10
Android 3.x (Honeycomb)	11-13 (tablets)
Android 4.0.x (Ice Cream Sandwich)	14, 15

API Versions

Android OS	API Level
Android 4.1, 4.2, 4.3 (Jelly Bean)	16, 17, 18
Android 4.4 (KitKat)	19
Android 4.4W (Wear)	20
Android 5.0, 5.1.x (Lollipop)	21, 22
Android 6.0 (Marshmallow)	23
Android 7.0 (Nougat)	24-25

API Versions

Android OS	API Level
Android 8.0 (Oreo)	26-27
Android 9.0 (Pie)	28
Android 10 (Quince)	29
Android 11 (Red Velvet)	30
Android 12 (SnowCone)	31/32

API Versions

Android OS	API Level
Android 13 (Tiramisu)	33
Android 14 (Upside Down Cake)	34
Android 15 (VanillaIceCream)	35
Android 16 (Baklava)	36

Android Platform/API Version Distribution

ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.4 KitKat	19	
5 Lollipop	21	99.9%
5.1 Lollipop	22	99.8%
6 Marshmallow	23	99.3%
7 Nougat	24	98.6%
7.1 Nougat	25	98.0%
8 Oreo	26	97.4%
8.1 Oreo	27	96.4%
9 Pie	28	93.4%
10 Q	29	87.6%
11 R	30	77.4%
12 S	31	61.5%
13 T	33	48.7%
14 U	34	31.9%
15 V	35	4.5%

Last updated: April 1, 2025

V

Camera and media

Low Light Boost
In-app Camera Controls
HDR headroom control
Loudness control
Virtual MIDI 2.0 Devices
More efficient AV1 software decoding

Connectivity

Satellite support
Smoother NFC experiences
Wallet role

Developer productivity and tools

OpenJDK 17 updates
PDF improvements
Automatic language switching refinements
Improved OpenType Variable Font API
Granular line break controls
App archiving

Graphics

Modernizing Android's GPU access
Improvements for Canvas

Large screens and form factors

Improved large screen multitasking
Cover screen support

<https://developer.android.com/about/versions/15>

Performance and battery

ApplicationStartInfo API
Detailed app size information
App-managed profiling
SQLite database improvements
Android Dynamic Performance
Framework updates

Privacy

Screen recording detection
Expanded IntentFilter capabilities
Private space
Query most-recent user selection for
Selected Photos Access
Privacy Sandbox on Android
Health Connect
Partial screen sharing

Security

Integrate Credential Manager with autofill
Integrate single tap sign-up and sign-in with biometric prompts
Key management for end-to-end encryption
Permission checks on content URIs

User experience and system UI

Richer widget previews with Generated Previews API
Picture-in-picture improvements
Improved Do Not Disturb rules

OK

Cancel

Android device distribution

Android Virtual Device (AVD)

Android Virtual Devices

- Emulators from Google
- External emulators - Genymotion

Characteristics

- Processor, screen, camera, memory, (RAM, internal, external), API version

Emulating

- ARM
- X86, x64
 - Needs Intel HAXM and virtualization support processor

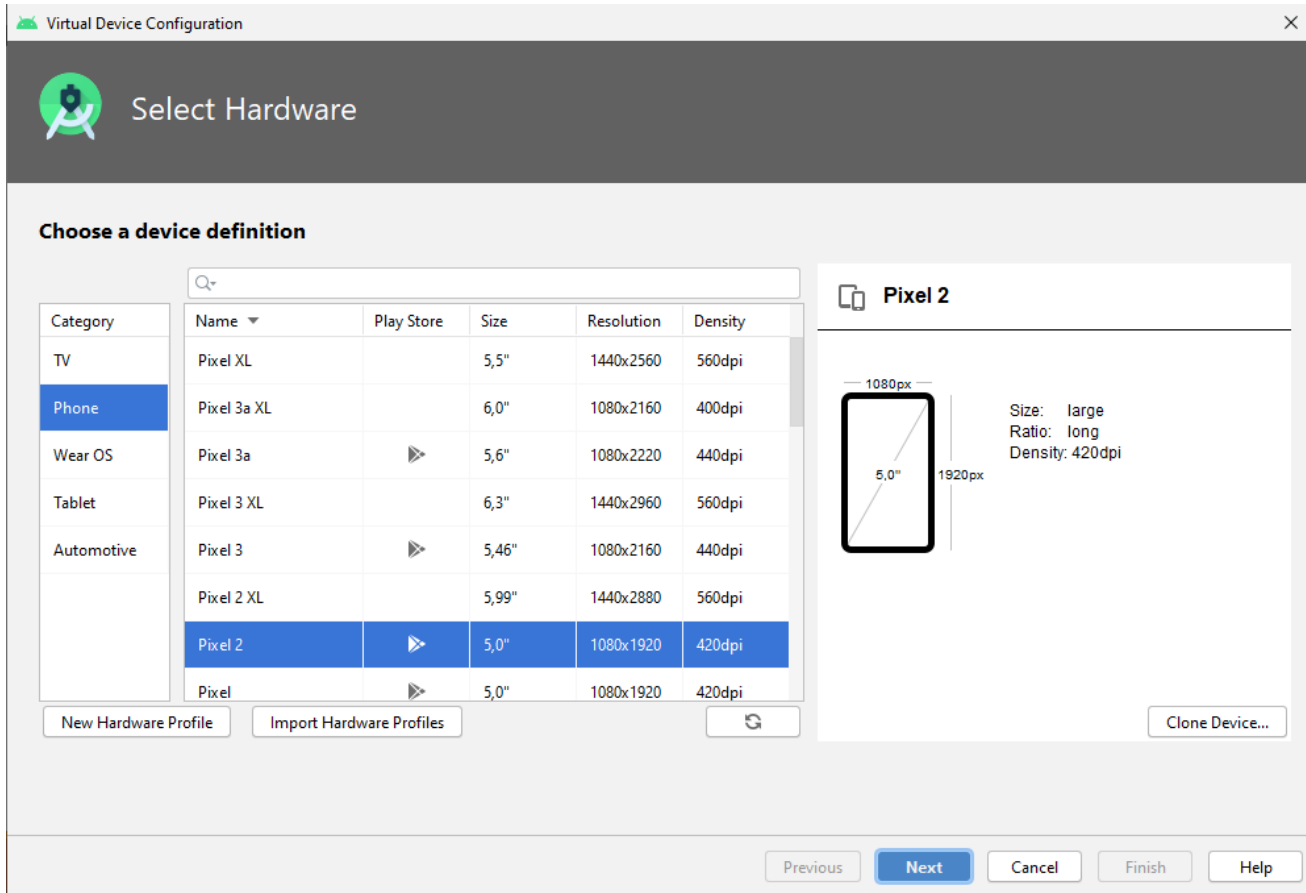
Communication through ADB

AVD Manager

Virtual devices management

Can be accessed directly or through the local IDE

Defines the internal characteristics of a virtual device



AVD Manager

Virtual Device Configuration



System Image

Select a system image

Recommended

x86 Images

Other Images

Release Name	API Level	ABI	Target
R Download	30	x86	Android 10.0+ (Google Play)
Q Download	29	x86	Android 10.0 (Google Play)
Pie Download	28	x86	Android 9.0 (Google Play)
Oreo Download	27	x86	Android 8.1 (Google Play)
Oreo Download	26	x86	Android 8.0 (Google Play)
Nougat	25	x86	Android 7.1.1 (Google Play)
Nougat Download	24	x86	Android 7.0 (Google Play)

Nougat



API Level

25

Android

7.1.1

Google Inc.

System Image

x86

We recommend these Google Play images because this device is compatible with Google Play.

Questions on API level?

Previous

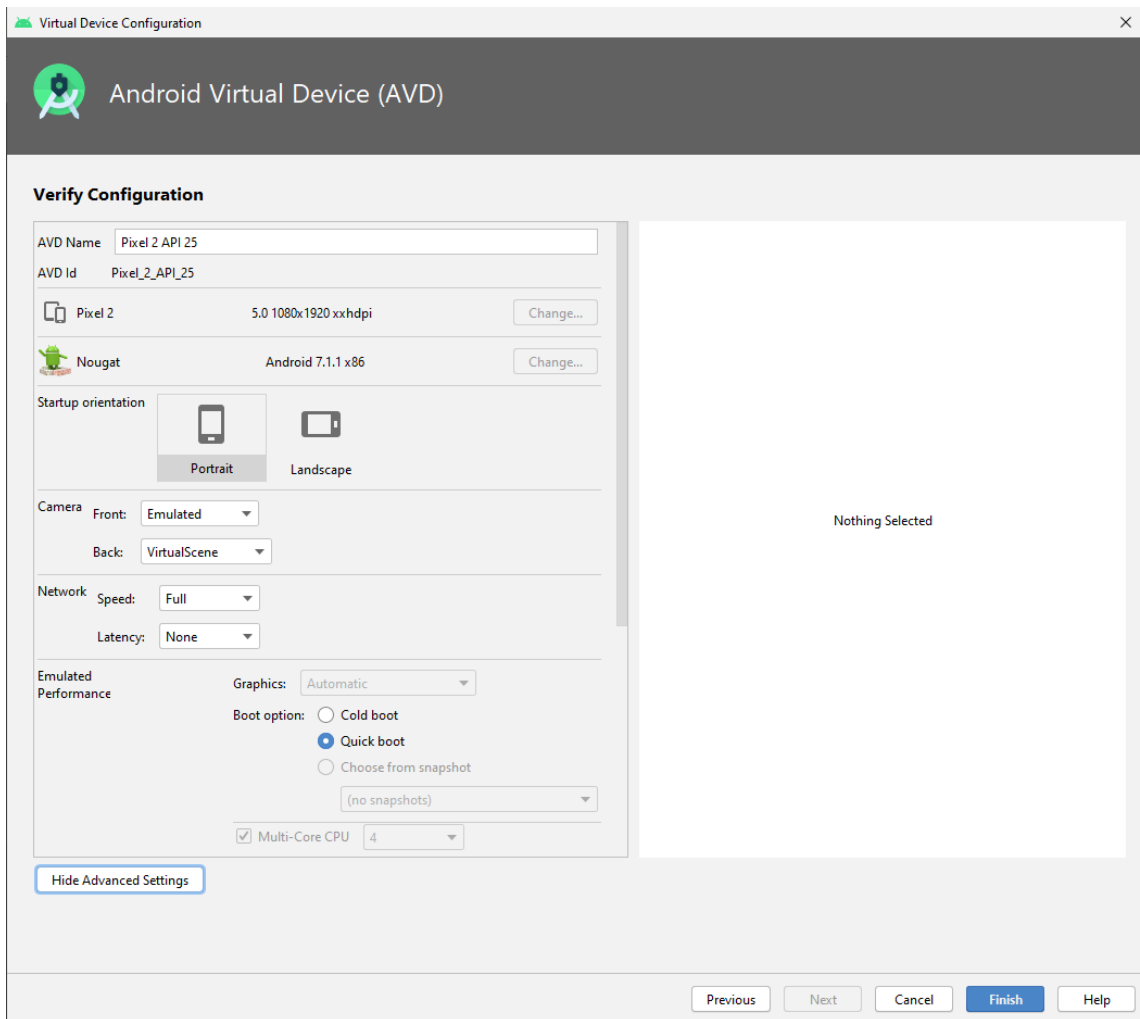
Next

Cancel

Finish

Help

AVD Manager



AVD Manager

Android Profiler

Gives access to devices

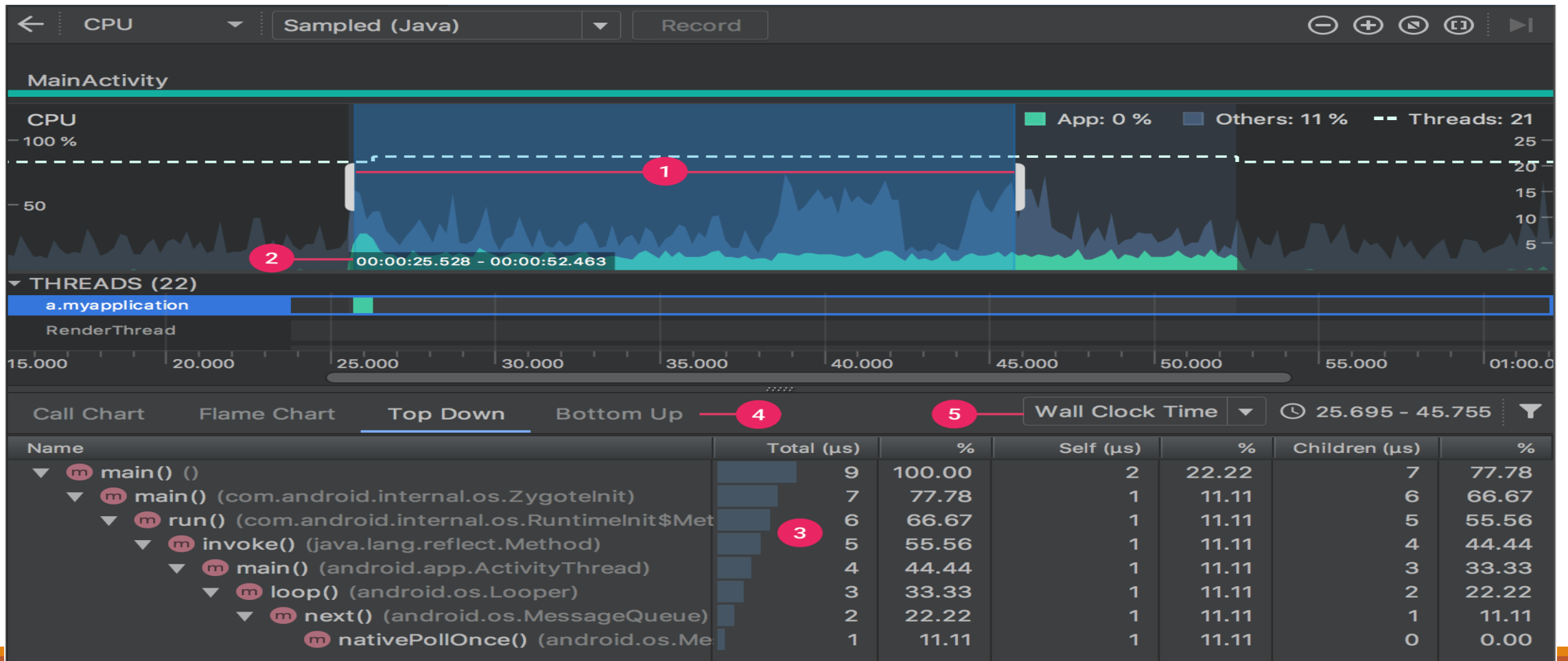
- virtual
- physical

Management

- Process
- Memory
- Network
- Messages (LogCat)
- Sensors
- File system

Access: **View > Tool Windows > Profiler**

Android Profiler



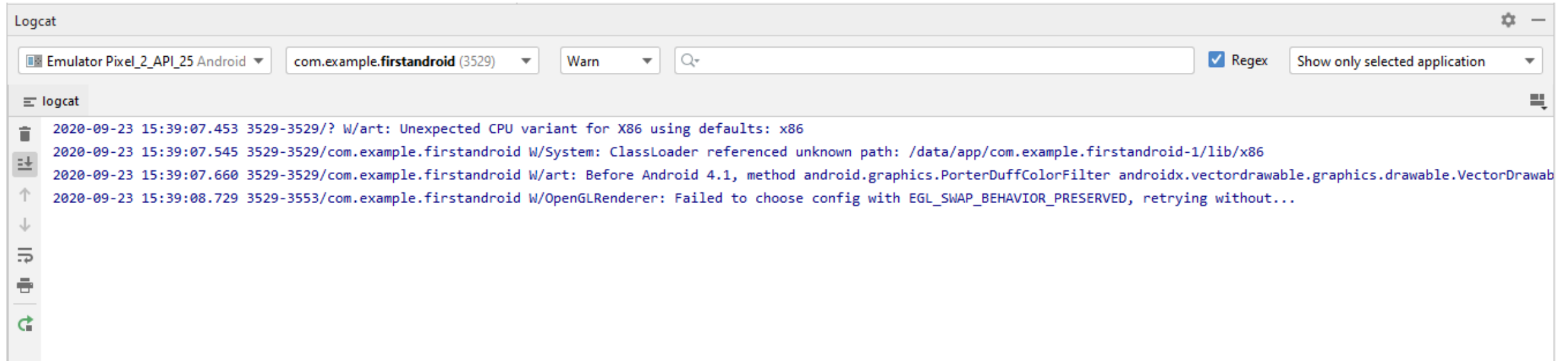
Message console (LogCat)

Displays messages sent by applications

Messages

- Warnings (w)
- Debug (d)
- Errors (e)
- Info (i)
- Detail info or verbose (v)
- Assert

Message console (LogCat)



Message console (LogCat)

Messages are displayed using the `android.util.Log` class

Each message type has its own static method:

- `e()`, `w()`, `i()`, `d()`, `v()`, `wtf()`

Parameters:

- Message source identifier (String)
 - Message tag for easily group messages
- Message Content (String)

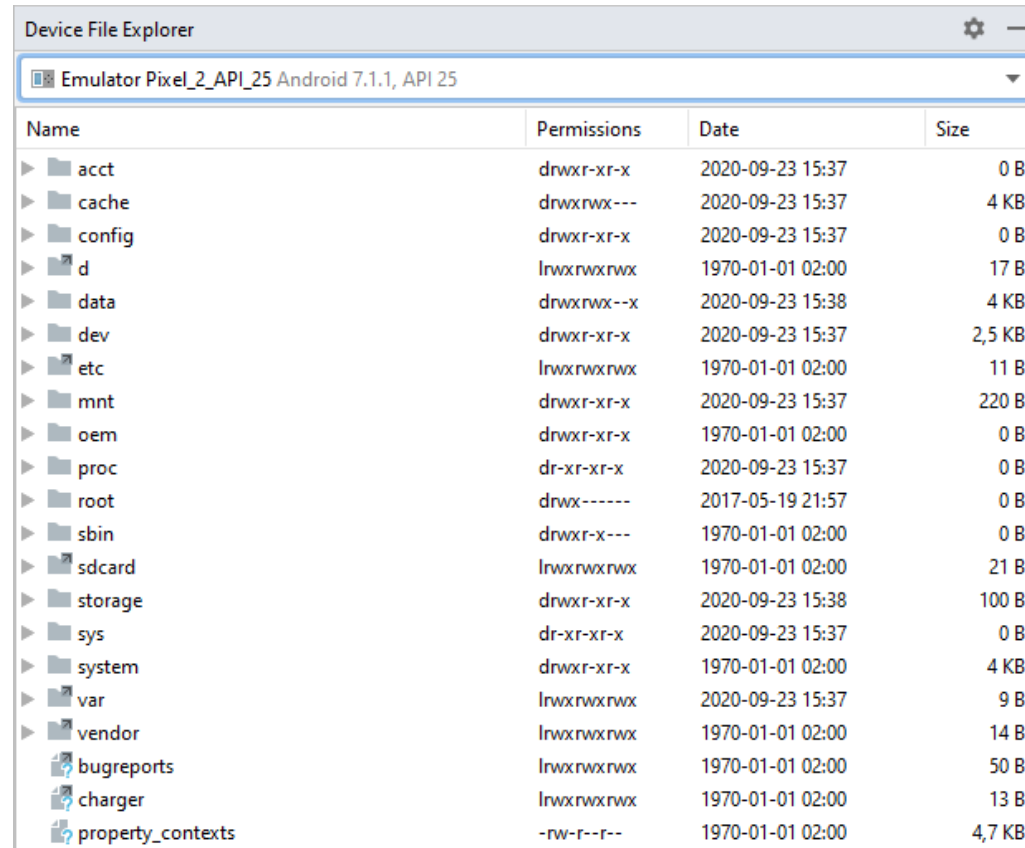
Generic static method

- `println()`
- The first parameter includes message type: `Log.ASSERT`, `Log.ERROR`, `Log.INFO` etc.

Example

- `Log.i("Activity", "Message content");`
- `println(Log.ASSERT, "Activity", "Message content");`

Device File Explorer



The screenshot shows the 'Device File Explorer' application window. At the top, there is a title bar with the text 'Device File Explorer' and a settings icon. Below the title bar is a dropdown menu showing 'Emulator Pixel_2_API_25 Android 7.1.1, API 25'. The main content area displays a table of files and directories. The table has four columns: 'Name', 'Permissions', 'Date', and 'Size'. The files are listed in a tree-like structure, with folders expanded to show their contents. The files are sorted by name, and the table includes icons for each file type (folders, files, and executables).

Name	Permissions	Date	Size
▶ acct	drwxr-xr-x	2020-09-23 15:37	0 B
▶ cache	drwxrwx---	2020-09-23 15:37	4 KB
▶ config	drwxr-xr-x	2020-09-23 15:37	0 B
▶ d	lrwxrwxrwx	1970-01-01 02:00	17 B
▶ data	drwxrwx--x	2020-09-23 15:38	4 KB
▶ dev	drwxr-xr-x	2020-09-23 15:37	2,5 KB
▶ etc	lrwxrwxrwx	1970-01-01 02:00	11 B
▶ mnt	drwxr-xr-x	2020-09-23 15:37	220 B
▶ oem	drwxr-xr-x	1970-01-01 02:00	0 B
▶ proc	dr-xr-xr-x	2020-09-23 15:37	0 B
▶ root	drwx-----	2017-05-19 21:57	0 B
▶ sbin	drwxr-x---	1970-01-01 02:00	0 B
▶ sdcard	lrwxrwxrwx	1970-01-01 02:00	21 B
▶ storage	drwxr-xr-x	2020-09-23 15:38	100 B
▶ sys	dr-xr-xr-x	2020-09-23 15:37	0 B
▶ system	drwxr-xr-x	1970-01-01 02:00	4 KB
▶ var	lrwxrwxrwx	2020-09-23 15:37	9 B
▶ vendor	lrwxrwxrwx	1970-01-01 02:00	14 B
bugreports	lrwxrwxrwx	1970-01-01 02:00	50 B
charger	lrwxrwxrwx	1970-01-01 02:00	13 B
property_contexts	-rw-r--r--	1970-01-01 02:00	4,7 KB

Resources

C. Boja, C. Ciurea, M. Doinea – Android Mobile Applications: A Practical Development Guide, ASE Press, 2015, 404 pp., ISBN 978-606-34-0033-9

S. Komatineni, D. MacLean – Pro Android 4, Apress, 2012

R. Meier – Professional Android 4 Application Development, Wiley, 2012

P. Pocatilu, Programarea dispozitivelor mobile, Editura ASE, 2012

P. Pocatilu, I. Ivan ș.a. – Programarea aplicațiilor Android, Editura, ASE, 2015

<http://developer.android.com>